從混沌到清晰：用 SRAF 賦能次世代情報分析

Structured Retrieval Agent Framework 技術解析與效益展示

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挑戰：我們正被非結構化資料淹沒

資訊過載 (Information Overload)：

每日報告、電報、摘要數量龐大（如本專案模擬的 30,000 份文件），人力難以完全消化。

查詢困難 (Difficult to Query)：

無法對文本進行精準、跨文件的複雜查詢（例如：「過去三個月某型號潛艦的活動趨勢?」）。

分析耗時 (Time-Consuming Analysis)：

人工匯總、趨勢分析與報告撰寫，耗費大量專家時間，從數小時到數天不等。

缺乏可追溯性 (Lack of Traceability)：

分析結論難以快速回溯至原始證據，導致可信度受質疑，無法作為決策依據。

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我們的願景：與你的資料庫直接對話

即時問答？

（用自然語言提問「最近 3 個月 095 核潛艦活動情況？」並獲得即時、精準的答案？）

自動生成報告？

（一鍵生成包含趨勢洞察與完整 SWOT 分析的深度報告？）

100% 可驗證？

（確保 AI 生成的每一條洞見，都有明確的原文證據支持，杜絕幻覺？）

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我們的解決方案：SRAF (Structured Retrieval Agent Framework)

一個將海量非結構化文本，轉化為可查詢、可分析、可追溯的**結構化知識庫**的 AI 框架。

LLM 驅動的資訊萃取
(LLM-Powered Information Extraction)

存入 Elasticsearch 知識庫
(Store in an Elasticsearch Knowledge Base)

AI 代理人查詢、推理與生成
(AI Agent for Query, Reasoning & Generation)

This process is a form of **Structured RAG (Retrieval-Augmented Generation).**

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技術核心：從文本到結構化資料的魔法

Input

114 年 01 月 01 日情勢摘要
今日 0824 起，空偵站掌握對岸活動跡象，約 0756 時雷達偵獲 052D 驅逐艦於東海一帶出現...

Engine

高度規則化的提詞工程
(Sophisticated Prompt Engineering)

- 日期與時間判讀 (核心規則)
- 民國年轉換
- 四位數的判讀優先序

Output

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{
  "article_title": "114 年 01 月 01
日情勢摘要",
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  "core_entities": ["052D 驅逐艦"],
  "locations": ["東海"],
  "Event_Timeline": [{
    "date_iso": "2025-01-01",
    "description": "約 0756 時雷達
偵獲 052D 驅逐艦..."
  }]
}
```

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實戰演練：SRAF 驅動的軍事情報分析代理人
Noto Sans TC

任務 (Mission) : 分析敵方潛艦動態

資料集 (Dataset): 30,000+ 篇模擬軍事情報文本 (情勢摘要、監偵回報)

目標 (Objective): 展示 SRAF 如何從海量文本中自動提煉可執行的洞見

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成果展示 (1): 一鍵生成深度策略報告

User Prompt

整理 095 新型核潛艦動態，產製 2000 字總結報告，包含趨勢與 SWOT 分析。

AI-Generated Report

STRATEGIC REPORT

Summary

活動頻度略增、地理分布廣泛、電子靜默與高速航跡...

Trend Analysis

偵測次數呈緩升，活動海域逐步擴大...

SWOT Analysis

S	W
隱蔽技術先進、遠程作戰能力強	在特定環境仍可能被雷達/電偵捕捉
O	**T**
我方可藉此大量累積反潛偵測資料	對手若結合更完整聲納網，區域衝突風險上升

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成果展示 (2)：每一個答案，都有證據支持

最近 3 個月 095 核潛艦活動情況？

在最近 3 個月內，資料庫未顯示 095 核潛艦的任何偵測或活動紀錄。最近的報導均為 2025 年 6 月的偵測，已超出查詢時間範圍。

Citations

Source Snippet 1

... 2025-06-23
... 約 0730 時雷達偵獲 **095 新型核潛艦** 於 **南海** 一帶出現...

Source Snippet 2

... 2025-06-24
... 約 0738 時雷達偵獲 **095 新型核潛艦** 於 **廣東沿岸**...

100% 可追溯性 (100% Traceability)

AI 的結論完全基於數據，杜絕「幻覺」

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核心效益：SRAF 為何是遊戲規則的改變者

效率躍升 (Efficiency Leap)

分析師從「搜尋與匯總」中解放，專注於「判斷與決策」。

分析時間從數小時縮短至數秒。

洞見深化 (Deeper Insights)

啟用趨勢分析、異常偵測等過去難以實現的複雜分析維度。

信任加固 (Reinforced Trust)

每項結論皆可追溯來源，建立人與 AI 之間可驗證的信任關係。

知識沉澱 (Knowledge Capital)

將瞬時的文本情報，轉化為永續、可擴展、可複利的結構化知識資產。

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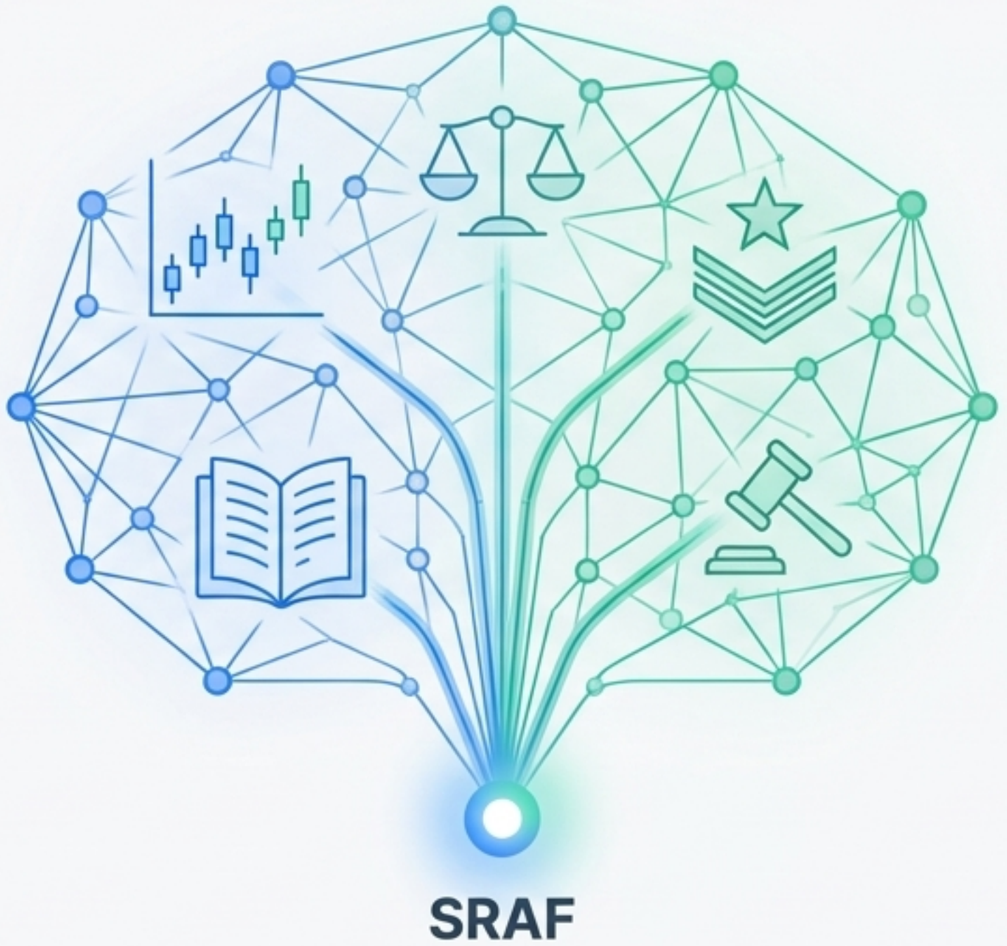
我們的下一步：賦能更多業務場景

採用 SRAF 框架，打造組織專屬的知識大腦

- 軍事情報分析
  (Military Intelligence Analysis)
- 金融市場輿情監測
  (Financial Market Intelligence)
- 法律文件審閱與判例查詢
  (Legal Document Review)
- 人權與公眾事務議題追蹤
  (Human Rights & Public Affairs Tracking)
- 學術研究文獻整理
  (Academic Literature Synthesis)



Let's build the future of intelligence, together.

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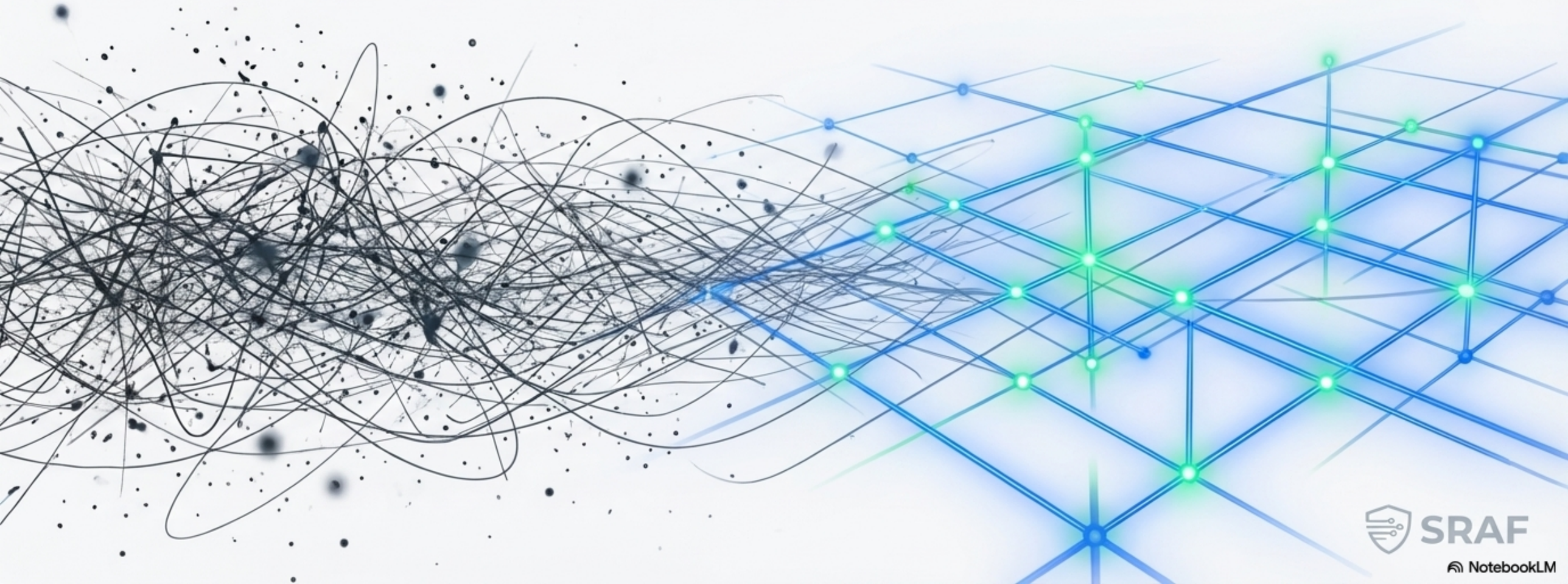
From Chaos to Clarity: Powering Next-Gen Intelligence Analysis with SRAF

A Technical Showcase of the Structured Retrieval Agent Framework



SRAF

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The Challenge: We're Drowning in Unstructured Data

Information Overload: Massive volumes of daily reports (like the 30,000 documents in our simulation) are impossible for humans to fully process.

Difficult to Query: Inability to perform precise, cross-document queries (e.g., "What's the activity trend of a specific submarine model in the last 3 months?").

Time-Consuming Analysis: Manual aggregation, trend analysis, and report writing consume expert hours that can take from hours to days.

Lack of Traceability: Analytical conclusions are difficult to trace back to original evidence, undermining credibility for decision-making.

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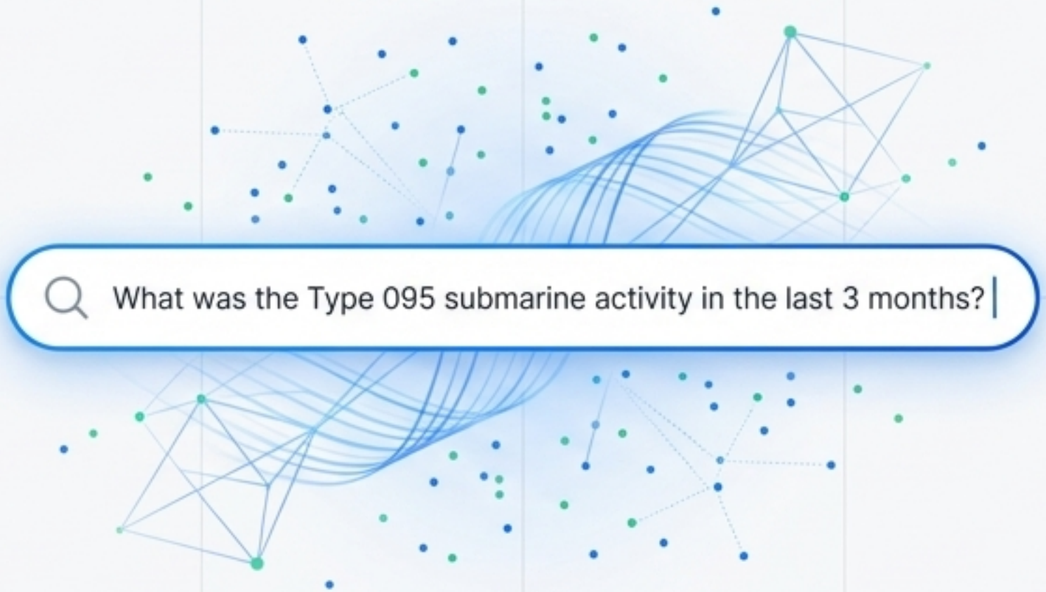
The Vision: A Direct Conversation with Your Data

Ask in real-time?

(Ask 'What was the Type 095 submarine activity in the last 3 months?' in plain language and get an instant, accurate answer?)



Generate reports automatically?

(Generate in-depth reports with trend insights and a full SWOT analysis with a single click?)

Verify every claim 100%?

(Ensure every AI-generated insight is backed by verifiable source citations, eliminating hallucinations?)

Our Solution: SRAF (Structured Retrieval Agent Framework)

An AI framework that transforms massive volumes of unstructured text into a queryable, analyzable, and traceable structured knowledge base.

LLM-Powered Information Extraction

Store in an Elasticsearch Knowledge Base

AI Agent for Query, Reasoning & Generation

This process is a form of **Structured RAG (Retrieval-Augmented Generation).**

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The Core Tech: The Magic of Turning Text into Structure

Input

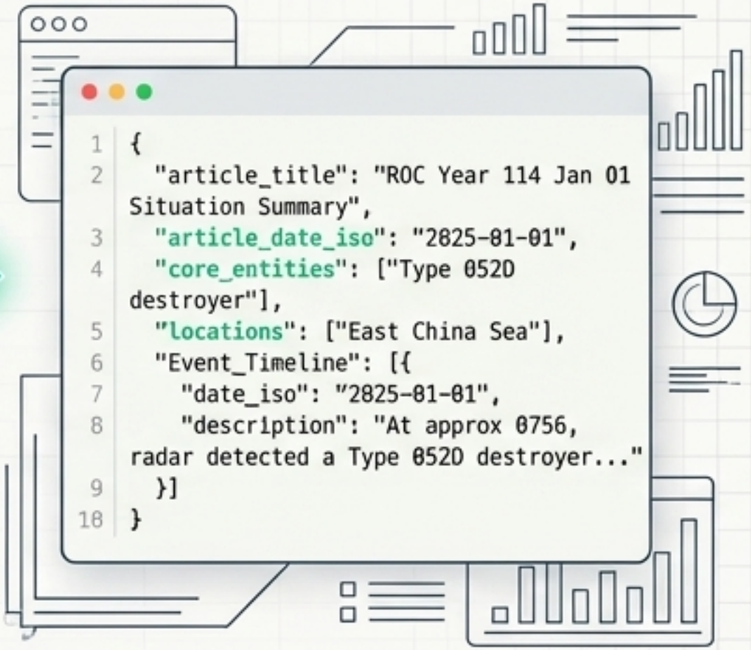
"ROC Year 114, Jan 01 Situation Summary... At approx 0756, radar detected a Type 052D destroyer near the East China Sea..."

Engine

Sophisticated Prompt Engineering

- Date & Time Parsing (Core Rules)
- ROC Year Conversion
- 4-Digit Number Parsing Priority

Output



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  "article_date_iso": "2825-01-01",
  "core_entities": ["Type 052D destroyer"],
  "locations": ["East China Sea"],
  "Event_Timeline": [{
    "date_iso": "2825-01-01",
    "description": "At approx 0756, radar detected a Type 052D destroyer..."
  }]
}
```

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SRAF in Action: The Military Intelligence Analysis Agent

30,000+

Documents Analyzed

Mission: Analyze adversary submarine dynamics.

Dataset: 30,000+ simulated military intelligence texts (situation reports, surveillance logs).

Objective: Demonstrate SRAF's ability to automatically distill actionable insights from massive text volumes.

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The Results (1): Deep Strategic Reports in a Single Click

User Prompt

Summarize the dynamics of the Type 095 submarine. Generate a 2000-word report incliding trends and a SWOT analysis.

AI-Generated Report

Summary

Slightly increased activity frequency, broad geographical distribution, electronic silence and high-speed maneuvers...

Trend Analysis

Detection counts show a gradual increase, with an expanding operational area...

SWOT Analysis

S - Strengths	W - Weaknesses
Advanced stealth tech, strong long-range capabilities.	Still detectable by radar/electronic surveillance in certain conditions.
O - Opportunities	**T - Threats**
Allows us to accumulate vast amounts of anti-submarine detection data.	Regional conflict risk increases if adversary integrates a more complete sonar network.

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The Results (2): Every Answer, Backed by Evidence

What was the Type 095 submarine activity in the last 3 months?

In the **last 3 months**, the database shows no detection or activity records for the Type 095 submarine. The most recent reports are from June 2025, which is outside the query time frame.

Citations

Source Snippet 1

... **2025-06-23**
... At approx 0730, radar detected a **Type 095 submarine** near the **South China Sea**...

Source Snippet 2

... **2025-06-24**
... At approx 0738, radar detected a **Type 095 submarine** off the **coast of Guangdong**...

Manrope Bold / Inter Regular

100% Traceability

AI conclusions are based purely on data, eliminating 'hallucinations'.

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The Impact: Why SRAF is a Game-Changer

Efficiency Leap

Frees analysts from “search and aggregate" to focus on "judge and decide.” **Reduces analysis time from hours to seconds.**

Deeper Insights

Enables complex analysis previously out of reach, like trend monitoring and anomaly detection.

Reinforced Trust

Builds human-AI trust with verifiable, source-backed conclusions.

Knowledge Capital

Transforms ephemeral text intelligence into a permanent, scalable, and compounding structured knowledge asset.

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Our Next Steps: Empowering More Business Domains

Adopt the SRAF framework to build your organization's specialized knowledge brain.

- Military Intelligence Analysis
- Financial Market Intelligence
- Legal Document Review & Case Law Search
- Human Rights & Public Affairs Tracking
- Academic Literature Synthesis

Let's build the future of intelligence, together.

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